Figure 1.

UNIVERSITY OF NAIROBI
FACULTY OF SCIENCE AND TECHNOLOGY
DEPARTMENT OF COMPUTING AND INFORMATICS

Fitness Misinformation Detection System Using Natural Language Processing

BY

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A project report submitted in partial fulfillment of the requirements for the award of Bachelor of Science in Computer Science of the University of Nairobi.

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DECLARATION

This proposal is entirely my own original work and has not been submitted, in whole or in part, for the award of any degree or diploma at this or any other University.

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Peter Kioko
1st October 2024

ABSTRACT

The rapid proliferation of health and wellness content on social media has fostered an environment where unverified exercise and nutrition claims spread unchecked, potentially leading to physical injuries and ineffective training practices among enthusiasts. While automated fact-checking tools exist, they predominantly focus on political discourse or general medical misinformation, leaving a critical gap in the detection of domain-specific fitness misinformation. This project proposes the development of an AI-Powered Fitness Misinformation Detection System designed to identify and verify such claims. The system will employ a three-tier architecture comprising a React-based web interface, a Python backend for Natural Language Processing (NLP), and a PostgreSQL database. By integrating OpenAI's Whisper API, the application will enable users to submit both text and audio content for analysis. Extracted claims will be cross-referenced against a curated dataset of peer-reviewed exercise science literature to generate validity confidence scores and evidence-based assessments. The successful implementation of this system aims to empower fitness enthusiasts with the tools necessary to distinguish evidence-based science from unfounded myths, ultimately contributing to safer wellness practices.

ABBREVIATIONS AND ACRONYMS

Abbreviation/Acronym	Definition
AI	Artificial Intelligence
API	Application Programming Interface
AWS	Amazon Web Services
BERT	Bidirectional Encoder Representations from Transformers
DFD	Data Flow Diagram
ERD	Entity-Relationship Diagram
ML	Machine Learning
MVP	Minimum Viable Product
NER	Named Entity Recognition
NLP	Natural Language Processing
REST	Representational State Transfer
TOR	Terms of Reference
UI	User Interface
UX	User Experience

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CHAPTER 1: INTRODUCTION

1.1 Background

The proliferation of social media platforms has transformed how health and wellness information is disseminated and consumed globally (González-Fernández et al., 2021). While this democratization has increased access to health knowledge, it has simultaneously created an environment where unverified and potentially harmful information can spread rapidly (Yeung et al., 2022). Health misinformation on social media platforms has become a significant public health concern, with systematic reviews indicating that misleading health content is prevalent across multiple digital platforms (Suarez-Lledo & Alvarez-Galvez, 2021).

Within the broader category of health misinformation, exercise and nutrition-related false claims represent a particularly concerning subset. Marocolo et al. (2021) examined Brazilian social media content and found widespread dissemination of exercise misinformation, while Denniss, Lindberg, and McNaughton (2023) identified significant quality issues in online nutrition information through systematic content analysis. The sports science community has increasingly called attention to baseless claims and pseudoscience in health and wellness spaces, emphasizing the urgent need for evidence-based counter-measures (Tiller, Sullivan, & Ekkekakis, 2023).

Current approaches to combating health misinformation rely heavily on manual expert review processes or general fact-checking systems that lack domain-specific knowledge (Swire-Thompson & Lazer, 2022). Automated detection systems for health misinformation represent an emerging research area, with recent studies exploring machine learning approaches for identifying false health claims in digital content (Di Sotto & Viviani, 2022; Schlicht et al., 2024). However, existing systems predominantly focus on medical misinformation rather than exercise and nutrition domains, creating a significant gap in automated verification capabilities for wellness-related content.

This project addresses the identified gap by developing an artificial intelligence system specifically designed to detect and verify claims related to exercise techniques, workout routines, and basic nutritional information. The focus on non-medical wellness content allows for meaningful impact while maintaining appropriate ethical boundaries regarding medical advice and intervention.

1.2 Problem Statement

Current digital platforms lack automated systems capable of identifying and verifying exercise and nutrition-related misinformation. This results in the unchecked spread of potentially harmful wellness advice that may lead to exercise-related injuries, unrealistic expectations, and adoption of unsafe practices among fitness enthusiasts and beginners.

1.3 Objectives for the Project

1. Develop a natural language processing system for extracting and classifying exercise and nutrition claims from textual content
2. Create a machine learning model trained on peer-reviewed exercise science literature to assess claim validity and provide confidence scores
3. Design and implement a web-based interface that allows users to submit content for automated fact-checking with appropriate disclaimers
4. Validate system accuracy through testing with known misinformation examples and expert evaluation

1.4 Scope

The scope of this project is to develop a Minimum Viable Product (MVP) of a fitness misinformation detection system that focuses on exercise techniques, workout routines, and basic nutritional claims for English-language content.

1.4.1 Functionality Scope

The system will support automated claim extraction from user-submitted text content, using NLP techniques to identify specific fitness and nutrition-related assertions. It will provide claim verification through comparison with a curated database of peer-reviewed exercise science research, generating confidence scores that indicate the likelihood of claim validity based on available evidence. Users will receive detailed verification results that include the extracted claims, confidence scores, supporting or contradicting evidence summaries, and source citations to relevant research.

The system will include a submission history feature allowing users to view their previous fact-checking requests and results. Appropriate disclaimers will be prominently displayed throughout the interface, clarifying that the system provides information rather than medical advice and recommending consultation with healthcare professionals for personalized health decisions.

Processing will be limited to text inputs under 1000 words to ensure real-time response capabilities, with the system focusing on English-language content initially. The evidence database will contain peer-reviewed research published within the last decade, focusing specifically on exercise science and sports nutrition domains.

Features explicitly excluded from this MVP include: real-time monitoring of social media platforms, automated content moderation or removal capabilities, analysis of video or audio content, personalized fitness advice or workout plan generation, direct medical diagnosis or treatment recommendations, and multi-language support beyond English.

1.4.2 Time Scope

The project's timeline is set for two three-month semesters, totaling eight months from initiation to deployment. This duration will be broken down into six sprints following an Agile methodology: Sprint 1 (Weeks 1-4) for research and dataset curation; Sprint 2 (Weeks 5-8) for NLP pipeline development and claim extraction; Sprint 3 (Weeks 9-12) for ML model training and validation; Sprint 4 (Weeks 13-16) for web interface development; Sprint 5 (Weeks 17-20) for system integration and testing; and Sprint 6 (Weeks 21-24) for final deployment and documentation.

1.5 Justification

Health misinformation represents a significant public health challenge, with the World Health Organization identifying it as a critical threat to global health outcomes (Do Nascimento et al., 2022). The specific domain of exercise and nutrition misinformation affects millions of individuals seeking wellness guidance through digital platforms, potentially leading to preventable injuries and ineffective practices (Krishna & Thompson, 2021).

Automated detection systems offer scalable solutions to address the volume and speed at which misinformation spreads online (Wang, Yin, & Argyris, 2020). By focusing on exercise and nutrition domains rather than medical diagnosis or treatment, this system operates within appropriate ethical boundaries while providing measurable public health benefits. The technical innovation contributes to the broader field of automated fact-checking and demonstrates advanced applications of natural language processing in health communication.

The system benefits multiple stakeholder groups: individuals seeking reliable wellness information gain access to evidence-based verification, fitness professionals can reference validated content, and public health organizations benefit from reduced misinformation-related health risks. The research contributes to the growing body of knowledge in digital health literacy and responsible artificial intelligence development.

1.6 Constraints

Development will occur on standard computing hardware with 8GB minimum RAM requirements, utilizing cloud infrastructure for deployment and scalability. The implementation will use Python-based machine learning frameworks including TensorFlow or PyTorch for model development, with PostgreSQL for database management and React.js for frontend development. Cloud deployment will utilize AWS or Google Cloud Platform services to ensure accessibility and performance.

The system must comply with data protection regulations for user input handling and adhere to academic research usage rights and copyright laws. Implementation will include appropriate disclaimers regarding system limitations and the need for professional medical consultation. The project will maintain clear boundaries between information provision and medical advice to ensure ethical compliance and user safety.

CHAPTER 2: REVIEW OF SIMILAR SYSTEMs

2.1 Health Misinformation Detection Systems

The landscape of automated health misinformation detection has evolved significantly over recent years, with researchers exploring various approaches to identify false health claims in digital content. Schlicht et al. (2024) conducted a systematic review of automatic health misinformation detection systems, identifying key methodological approaches and performance metrics across existing studies. Their analysis revealed that most systems focus on general medical misinformation rather than specific domains such as exercise or nutrition, highlighting a significant gap that this project aims to address.

Di Sotto and Viviani (2022) developed a comprehensive framework for health misinformation detection in social web environments, emphasizing the importance of multi-modal approaches that consider both textual content and user behavior patterns. Their Vec4Cred model demonstrated that combining content analysis with credibility signals can significantly improve detection accuracy. However, their work highlighted the challenge of balancing detection accuracy with false positive rates, particularly in health domains where incorrect classifications can have serious consequences.

Recent advances in machine learning for health misinformation detection have shown promising results. Wang, Yin, and Argyris (2020) demonstrated the effectiveness of multimodal deep learning approaches for detecting medical misinformation on social media platforms, achieving significant improvements over traditional text-only classification methods. Their system incorporated both textual content and visual information, achieving accuracy rates of 82-85% on test datasets. However, their focus remained on general medical content rather than specialized wellness domains, and the computational requirements of their multimodal approach exceeded the resources available for many research projects.

Zhao, Da, and Yan (2021) explored the incorporation of behavioral features into machine learning approaches for detecting health misinformation in online communities. Their research demonstrated that user interaction patterns, posting frequency, and network analysis could complement content-based detection methods. While behaviorally-augmented models showed improved performance, they also raised privacy concerns and required access to platform-level data not always available to external researchers.

Barve and Saini (2021) proposed a novel hybrid approach to healthcare misinformation detection and fact-checking, incorporating both automated detection and human verification components. Their methodology addressed limitations of purely automated systems while maintaining scalability for large-volume content processing. The hybrid model achieved 78% accuracy in initial testing, with the human verification component catching errors that automated systems missed. However, the reliance on human reviewers created bottlenecks that limited real-time processing capabilities.

2.2 Natural Language Processing in Health Communication

The application of natural language processing techniques to health communication has gained considerable attention from researchers seeking to understand and improve health information quality online. Peng, Lim, and Meng (2023) conducted a systematic review of persuasive strategies in online health misinformation, revealing common linguistic patterns and rhetorical techniques used in false health claims. Their analysis identified six primary persuasive strategies: appeal to emotion, false authority citations, cherry-picked evidence, logical fallacies, testimonial manipulation, and scientific-sounding jargon without substance.

These linguistic patterns provide valuable features for automated detection systems. For example, misinformation content often uses more emotional language, makes more absolute claims (using words like "always" or "never"), and presents anecdotal evidence as scientific proof. Understanding these patterns helps inform the development of NLP pipelines that can extract and analyze claims more effectively.

Advanced language models, particularly transformer-based architectures like BERT (Bidirectional Encoder Representations from Transformers), have shown remarkable effectiveness in medical text understanding and classification tasks (Upadhyay, Pasi, & Viviani, 2023). These models demonstrate capability in capturing contextual nuances that are critical for distinguishing between legitimate health advice and misinformation. The bidirectional nature of BERT allows it to understand context from both directions in a sentence, which is particularly valuable when analyzing complex scientific claims where meaning depends heavily on qualifiers and conditions.

However, domain-specific fine-tuning remains necessary for optimal performance in specialized areas such as exercise science. Pre-trained language models learn general linguistic patterns from massive text corpora, but they may lack the specialized vocabulary and contextual understanding needed for fitness and nutrition domains. Fine-tuning these models on curated datasets of exercise science literature can significantly improve their performance on domain-specific tasks.

The challenge of claim extraction—identifying specific factual assertions within longer texts—has been addressed through various NLP techniques. Named Entity Recognition (NER) systems can identify fitness-related concepts, while dependency parsing can reveal relationships between entities that constitute claims. Recent research has explored the use of sequence-to-sequence models for claim extraction, treating it as a text summarization task where the goal is to extract verifiable factual statements from longer passages.

2.3 Exercise Science and Digital Health Information

The intersection of exercise science and digital health information represents an underexplored area in misinformation research. Marocolo et al. (2021) conducted one of the first systematic examinations of exercise misinformation on social media, focusing on Brazilian Portuguese content. Their study identified several common categories of false claims including unsafe exercise techniques, unproven performance enhancement methods, misleading nutritional advice, and pseudoscientific recovery protocols.

The researchers found that exercise misinformation often exploits legitimate scientific concepts but distorts them beyond recognition. For example, claims about "muscle confusion" exaggerate the principle of exercise variation, while "fat-burning zones" misrepresent the more nuanced relationship between exercise intensity and energy substrate utilization. These distortions are particularly problematic because they contain enough scientific-sounding language to seem credible to non-experts.

Denniss, Lindberg, and McNaughton (2023) performed a systematic review of online nutrition information quality, revealing significant variability in accuracy and evidence-based content across digital platforms. Their analysis of 127 nutrition-related websites and social media accounts found that only 31% provided information consistent with current evidence-based guidelines. Common problems included oversimplification of complex nutritional science, promotion of restrictive diets without scientific support, and conflation of correlation with causation in nutritional research.

The sports science community has increasingly recognized the proliferation of pseudoscientific claims in fitness and wellness spaces. Tiller, Sullivan, and Ekkekakis (2023) issued a call to action for the exercise science community to actively combat baseless claims and promote evidence-based practices through digital platforms and educational initiatives. They identified several factors contributing to the spread of fitness misinformation: the complexity of exercise science makes it difficult for lay audiences to evaluate claims critically; personal testimonials and before-after photos are more compelling than statistical evidence for many audiences; the fitness industry's commercial interests incentivize sensational claims over accurate information; and social media algorithms amplify engaging content regardless of its accuracy.

The economic incentives for fitness misinformation are substantial. Influencers can monetize large followings built on sensational claims, supplement companies benefit from pseudoscientific product endorsements, and content creators gain engagement through controversial or provocative fitness advice. These economic factors create a systematic bias toward misinformation in fitness content, making automated detection tools increasingly necessary.

2.4 Existing Fact-Checking Platforms and Tools

Several general-purpose fact-checking platforms have emerged to combat misinformation across various domains. Snopes, FactCheck.org, and PolitiFact have established reputations for manual fact-checking but focus primarily on political and general news claims rather than specialized health or fitness content. These platforms rely on expert journalists and researchers to verify claims through manual investigation—a process that is thorough but not scalable to the volume of fitness content being generated daily.

Health Navigator and HealthLine represent attempts to provide reliable health information online, but these platforms focus on providing original content rather than verifying external claims. While they serve an important educational function, they don't address the problem of misinformation spreading through social media and other platforms.

Google's Health Knowledge Graph and similar initiatives attempt to surface authoritative health information in search results, but these systems don't actively identify or flag misinformation. They work by promoting credible sources rather than detecting and labeling false claims, which means users must proactively seek out accurate information rather than being warned when they encounter misinformation.

ClaimBuster, developed by researchers at the University of Texas at Arlington, represents a more advanced approach to automated fact-checking. The system uses natural language processing to identify checkable factual claims in political discourse and prioritizes them for human fact-checkers. However, the system focuses on political statements rather than health or fitness content, and it still requires human verification rather than providing automated verification.

Full Fact, a UK-based charity, has developed automated tools to assist human fact-checkers by identifying claims that have been previously fact-checked and matching new claims to existing verifications. This approach reduces redundant work but doesn't solve the fundamental scalability problem of manual fact-checking.

In the fitness domain specifically, no comprehensive automated fact-checking tools currently exist. Some fitness professionals maintain blogs or social media accounts dedicated to debunking fitness myths, but these efforts are manual, inconsistent, and reach limited audiences compared to the spread of misinformation. The absence of specialized tools for fitness misinformation detection represents both a significant gap and an opportunity for this project.

2.5 Literature Review

2.5.1 The Global Context: Rise and Limitations of Health Tech Systems

The global landscape of health technology has expanded dramatically over the past decade, with digital health interventions becoming increasingly sophisticated and widespread. However, as Krishna and Thompson (2021) note in their comprehensive review of health communication and misinformation scholarship, the rapid digitization of health information has created new challenges alongside opportunities. The democratization of health information through digital platforms has empowered patients and health seekers, but it has also created pathways for misinformation to reach vulnerable populations.

Yeung et al. (2022) conducted a bibliometric study of medical and health-related misinformation on social media, analyzing 1,225 publications from 2012 to 2021. Their analysis revealed exponential growth in research attention to health misinformation, particularly following the COVID-19 pandemic. The study identified several key themes in the literature: the mechanisms by which health misinformation spreads on social platforms, the psychological factors that make people susceptible to false health claims, the public health consequences of widespread misinformation, and the technical approaches to detecting and countering false information.

The technical sophistication of health misinformation detection systems has improved significantly, but challenges remain. Fridman, Johnson, and Elston Lafata (2023) propose a comprehensive framework for health information and misinformation that distinguishes between different types of false claims: completely fabricated information, information that distorts or exaggerates legitimate research findings, information that is outdated based on newer evidence, and information that applies legitimate science inappropriately to different contexts.

This taxonomy is particularly relevant for fitness misinformation, where claims often fall into the second and fourth categories. For example, the concept of "spot reduction" (the idea that exercising specific body parts will preferentially reduce fat in those areas) distorts the legitimate observation that exercise can reshape body composition, while ignoring the systemic nature of fat metabolism.

Global health technology initiatives have focused primarily on medical information and disease management, creating a gap in wellness and preventive health domains. This gap is especially pronounced in developing regions where health technology research and development have historically been concentrated in high-income countries. The fitness misinformation detection system proposed in this project addresses this gap by focusing on preventive health information and developing a solution within an African academic context.

2.5.2 The Kenyan Market Gap: Digital Health Literacy Challenges

The Kenyan digital landscape presents unique challenges and opportunities for addressing health misinformation. Wasserman and Madrid-Morales (2019) conducted an exploratory study of "fake news" and media trust in Kenya, Nigeria, and South Africa, revealing that 88% of Kenyan respondents encountered misinformation online at least occasionally, with 30% reporting frequent exposure. Importantly, their research found that while Kenyans express concern about misinformation, many lack the critical evaluation skills needed to distinguish accurate information from false claims.

Tully (2022) examined responses to misinformation in the Kenyan context, identifying several factors that complicate efforts to combat false information. Trust in traditional authorities and institutions affects how people evaluate information credibility. The rapid adoption of social media has outpaced development of digital literacy education. Economic pressures create incentives for content creators to prioritize engagement over accuracy. Cultural factors influence which types of claims are accepted without scrutiny.

These challenges are particularly relevant for fitness misinformation. Wellness advice often comes from social media influencers whose credibility is based on physical appearance and personal charisma rather than scientific expertise. The aspirational nature of fitness content makes people more willing to believe claims that promise rapid results or easy solutions to complex problems.

The growing middle class in Kenya has shown increased interest in fitness and wellness, creating both demand for quality information and vulnerability to misinformation. With gym memberships and fitness classes becoming more accessible in urban areas, more Kenyans are seeking guidance on exercise techniques, workout programming, and nutrition

CHAPTER 3: SYSTEM ANALYSIS AND DESIGN

3.1 System Development Methodology

This project will adopt an Agile development methodology, specifically a modified Scrum approach tailored for solo academic development. The methodology emphasizes iterative development, continuous feedback, and adaptive planning to ensure the final product meets both technical requirements and user needs effectively.

3.1.1 Methodology Justification

The choice of Agile methodology is driven by several project-specific factors. First, uncertainty in model performance requires that machine learning components undergo iterative testing and refinement, as initial accuracy metrics may necessitate architectural adjustments. Second, technology integration complexity demands flexible planning when combining multiple technologies including OpenAI Whisper API, NLP libraries, machine learning frameworks, and web development stacks. Third, research-driven development means that literature review findings may influence technical decisions throughout the project lifecycle. Finally, the two-semester timeline necessitates efficient resource allocation with clear milestone deliverables for academic assessment.

3.1.2 Sprint Structure

The project timeline is organized into six development sprints, each approximately four weeks in duration:

Sprint 1 (Weeks 1-4): Project foundation including literature review completion, research database curation, development environment setup, and initial technology stack validation.

Sprint 2 (Weeks 5-8): Core NLP pipeline development focusing on text preprocessing, claim extraction algorithms, and entity recognition for fitness-related concepts.

Sprint 3 (Weeks 9-12): Machine learning model development including training data preparation, model selection and training, validation testing, and confidence scoring mechanism implementation.

Sprint 4 (Weeks 13-16): OpenAI Whisper integration for audio transcription, error handling for API failures, and audio-to-text verification workflow development.

Sprint 5 (Weeks 17-20): Web interface development with React frontend, REST API implementation, database integration, and user authentication setup.

Sprint 6 (Weeks 21-24): System integration testing, user acceptance testing with target demographic, bug fixes and refinements, and final documentation preparation.

3.2 System Analysis

3.2.1 Stakeholder Analysis

The project's success depends on meeting the needs of multiple stakeholder groups with distinct interests and requirements.

Primary Stakeholders include **fitness enthusiasts and beginners** who represent the core end-users seeking reliable information to guide their training and nutrition decisions. These users will provide direct feedback during testing phases and represent the primary value recipients of the system.

Secondary Stakeholders encompass **fitness professionals and personal trainers** who may use the system to verify claims encountered in their practice or to educate clients. **Health educators and researchers** could leverage the system as an educational tool or for analyzing misinformation trends in fitness domains. **Academic supervisors and evaluators** assess the project's technical merit and contribution to knowledge. **Exercise science experts** provide domain knowledge validation for claim categories and verification logic.

3.2.2 Current System Analysis

Analysis of the existing landscape reveals significant gaps that justify this project's development. **General fact-checking platforms** such as Snopes and FactCheck.org focus on political content and general news claims, lacking specialization in fitness and exercise science domains. **Health information portals** like HealthLine provide original educational content but do not offer automated verification of external claims found on social media. **Academic databases** including PubMed and Google Scholar provide access to research but require significant expertise to search effectively and synthesize findings—a barrier for general fitness enthusiasts.

Social media platforms employ content moderation systems primarily targeting hate speech and extreme content, with limited capability to identify subtle health misinformation. **Existing AI chatbots** can answer fitness questions but often lack rigorous fact-checking against current scientific literature and may propagate common misconceptions. This gap creates the need for a specialized system combining NLP claim extraction, domain-specific knowledge bases, and evidence-based verification mechanisms.

3.2.3 Functional Requirements

The system must deliver specific capabilities to fulfill its intended purpose:

FR1: Text Input Processing - The system shall accept text input up to 1000 words, perform preprocessing including tokenization and normalization, and handle both formal and informal language styles common in fitness content.

FR2: Audio Input Processing - The system shall accept audio file uploads in common formats (MP3, WAV, M4A) up to 10MB, integrate with OpenAI Whisper API for transcription, display transcribed text for user verification, and handle transcription errors gracefully with user feedback mechanisms.

FR3: Claim Extraction Module - The system shall identify fitness and nutrition-related assertions within submitted content, classify claims by category (exercise technique, nutrition advice, performance claims, recovery practices), extract relevant context surrounding each claim, and assign preliminary confidence indicators based on claim specificity.

FR4: Verification Engine - The system shall query the research database for relevant peer-reviewed literature, apply machine learning models to assess claim validity, generate confidence scores on a standardized scale (0-100%), identify supporting or contradicting evidence, and compile source citations with DOI links where available.

FR5: Results Presentation - The system shall display extracted claims with individual verification results, present confidence scores with visual indicators (color-coded badges), provide evidence summaries with direct quotes from research, include clickable citations to original sources, and offer export functionality for results as PDF reports.

FR6: User Account Management - The system shall support user registration and authentication, maintain submission history for registered users, allow users to bookmark or save verification results, and track usage metrics for system improvement.

FR7: Feedback Collection - The system shall enable users to rate verification accuracy, provide comments on results quality, flag incorrect classifications for review, and contribute to system improvement through structured feedback forms.

3.2.4 Non-functional Requirements

These requirements define the quality attributes determining system effectiveness and user satisfaction:

NFR1: Performance - The system must achieve text processing and claim extraction within 10 seconds for typical inputs, complete verification queries within 30 seconds for standard claim volumes, maintain audio transcription processing times under 2 minutes for 5-minute audio clips, and support concurrent processing of multiple user requests.

NFR2: Usability - The interface must feature intuitive navigation requiring minimal training, achieve task completion for core functions within 3 clicks or taps, maintain consistent design language across all screens, provide clear error messages with actionable guidance, and ensure accessibility compliance with WCAG 2.1 Level AA standards.

NFR3: Reliability - The system must target 95% uptime during active development and testing phases, implement robust error handling for API failures with graceful degradation, automatically retry failed operations with exponential backoff, maintain data integrity through transaction management, and provide system status indicators to users.

NFR4: Security - User authentication must employ industry-standard practices (bcrypt password hashing, JWT tokens), data transmission must use HTTPS encryption, API keys must be stored securely in environment variables, user-submitted content must be sanitized to prevent injection attacks, and privacy compliance must align with GDPR principles for data handling.

NFR5: Scalability - The architecture must support horizontal scaling of web services, implement database indexing for efficient research literature queries, use caching mechanisms (Redis) for frequently accessed data, optimize ML model inference for response time, and design API rate limiting to prevent abuse.

NFR6: Maintainability - Code must follow PEP 8 style guidelines for Python, implement comprehensive logging for debugging purposes, include inline documentation for complex algorithms, maintain test coverage above 70% for critical functions, and provide clear deployment documentation for future maintenance.

The system requires capabilities for natural language processing, machine learning inference, database management, and web-based user interaction. Functional requirements include text preprocessing, claim extraction, evidence retrieval, confidence scoring, and result presentation. Non-functional requirements specify response times under 10 seconds, system availability of 99% uptime, and support for concurrent user access.

3.3 System Design

3.3.1 Architectural Design

The proposed system adopts a three-tier architecture pattern, separating presentation, business logic, and data management concerns. This architectural approach ensures maintainability, scalability, and clear separation of responsibilities.

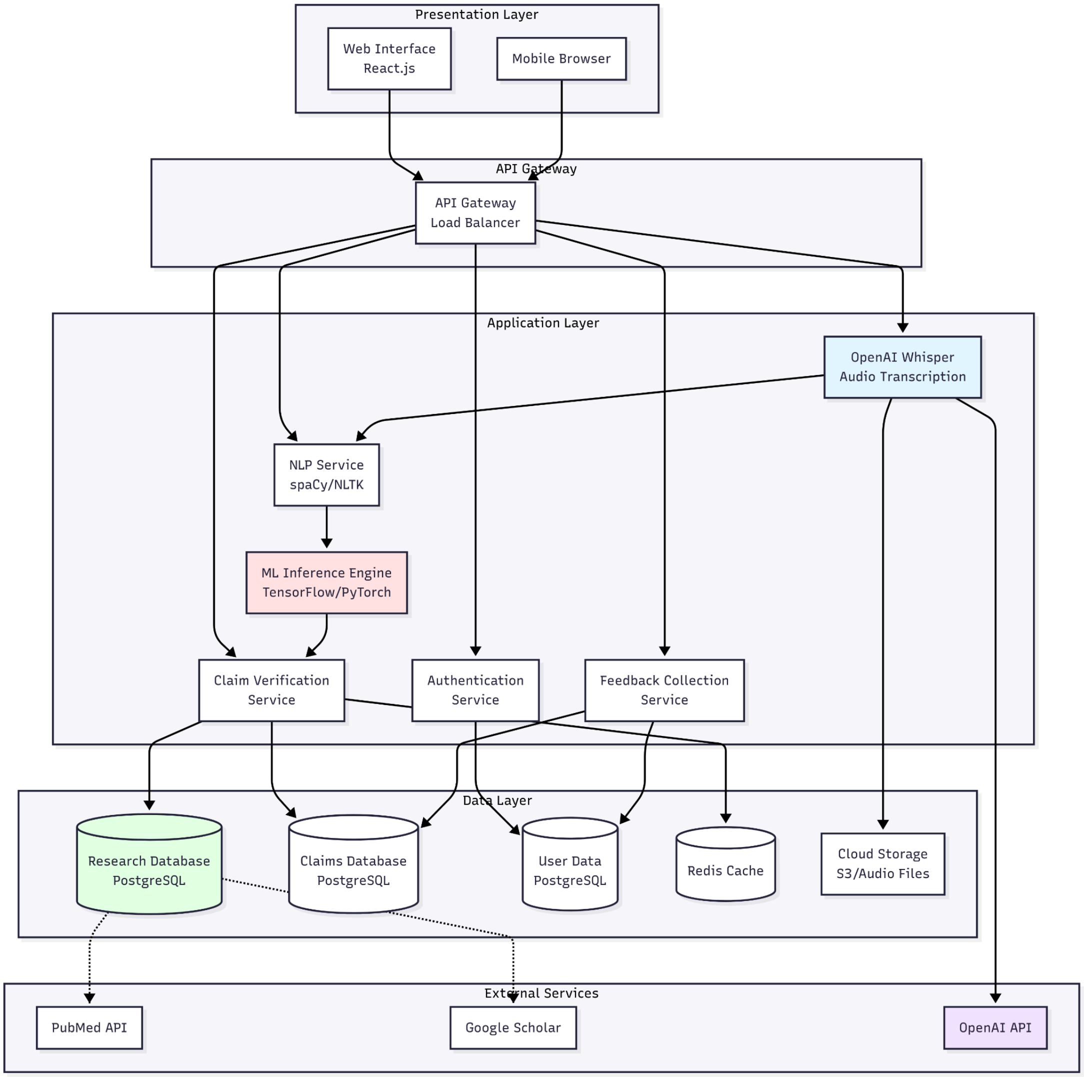

Figure 2.

3.3.2 Process Design

The main processing flow begins with user content submission, followed by text preprocessing and tokenization. Natural language processing services extract potential claims, which are then classified and structured. Machine learning models assess claims against the research database, generating confidence scores and supporting evidence citations before result presentation.

3.3.3 Database Design

The database schema will include tables for research papers, extracted claims, user queries, and system feedback. Research papers will store bibliographic information, abstracts, and full text content with appropriate indexing for efficient retrieval. Claims database will maintain structured representations of exercise and nutrition assertions with validity assessments and supporting evidence links

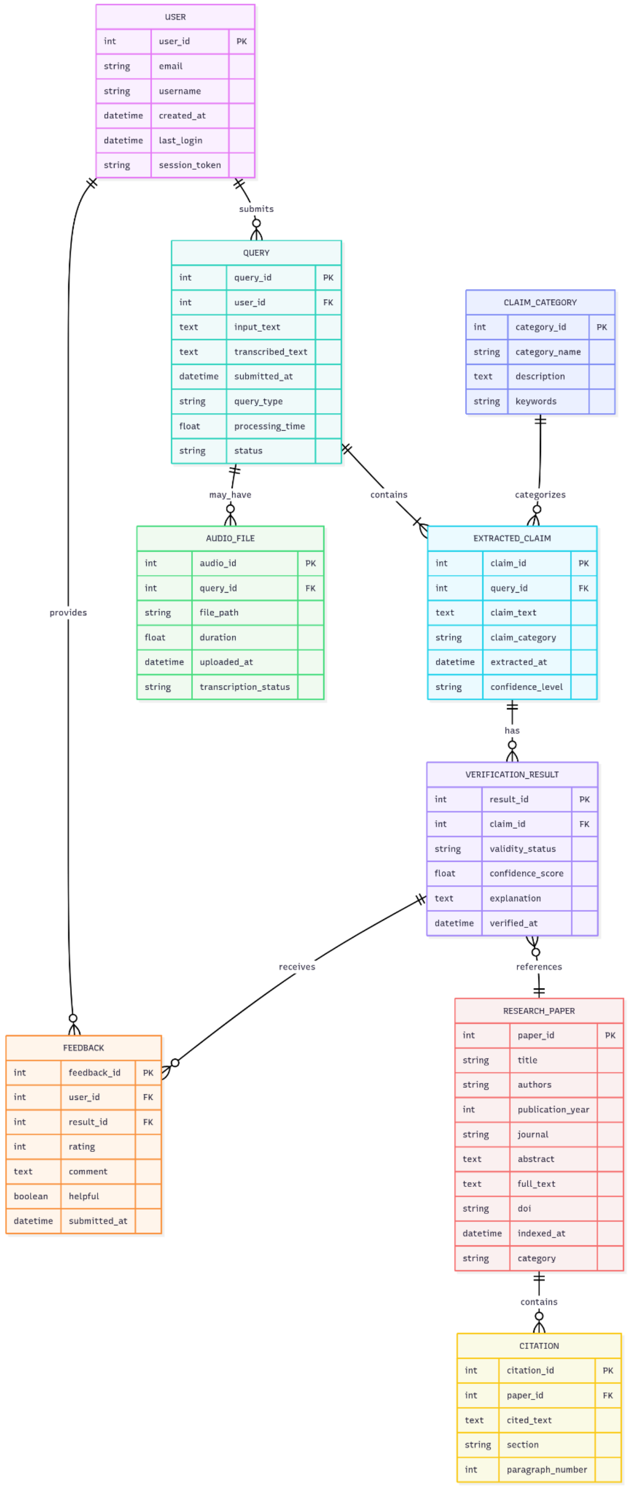
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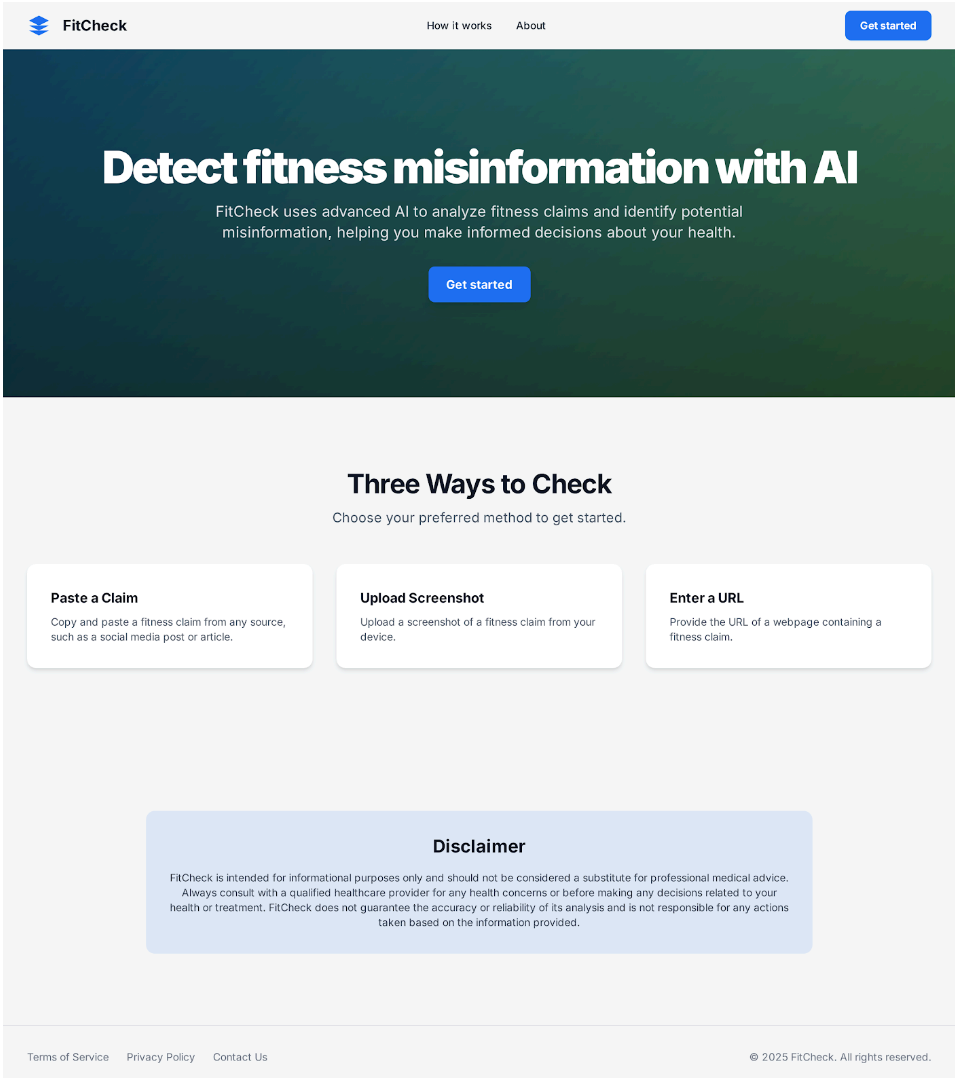
Figure 3.

3.3.4 User Interface Designs



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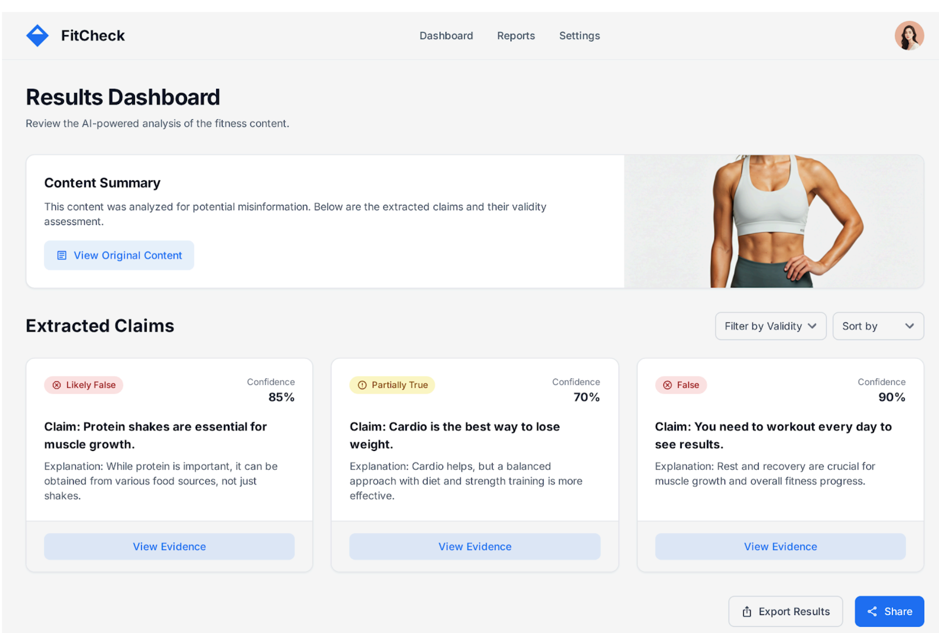


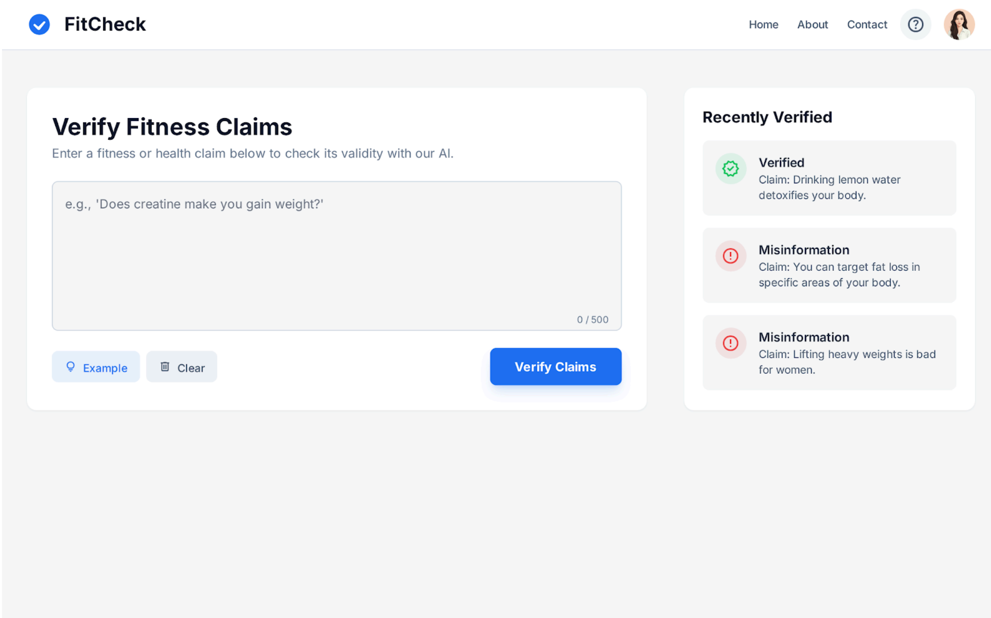
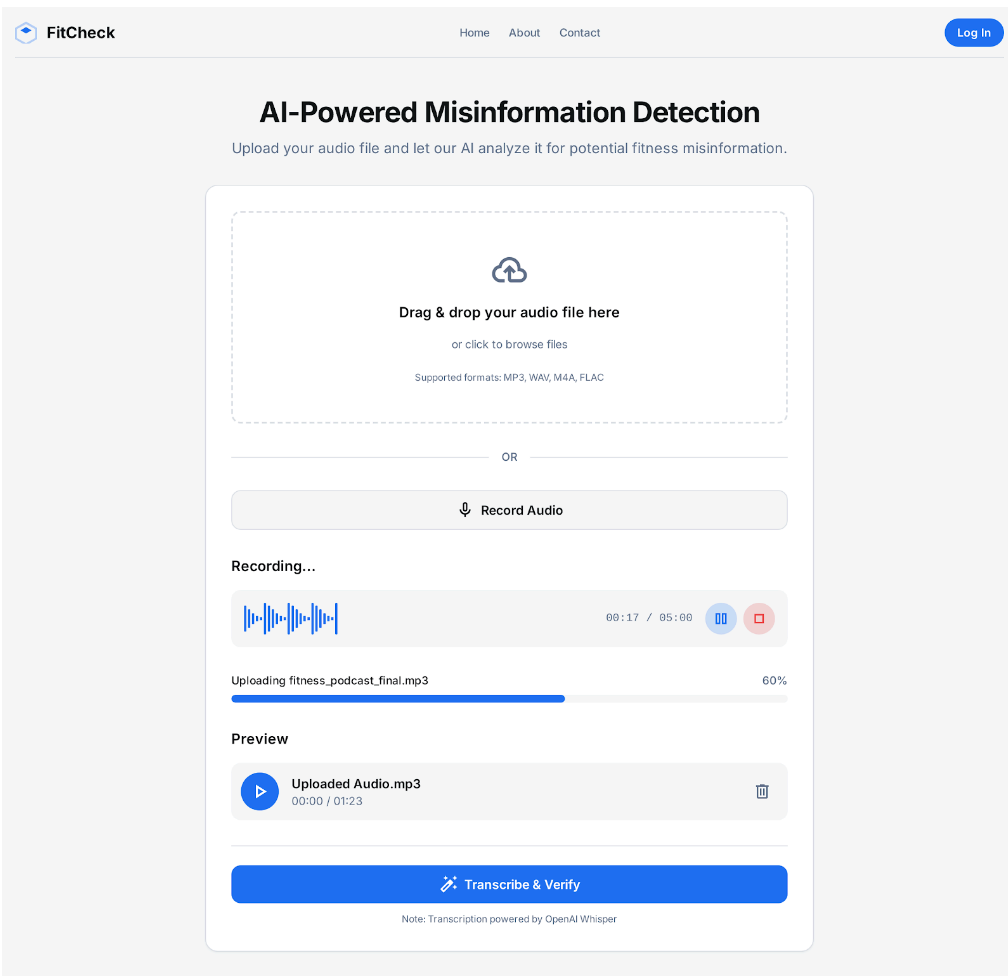


Figure 6.



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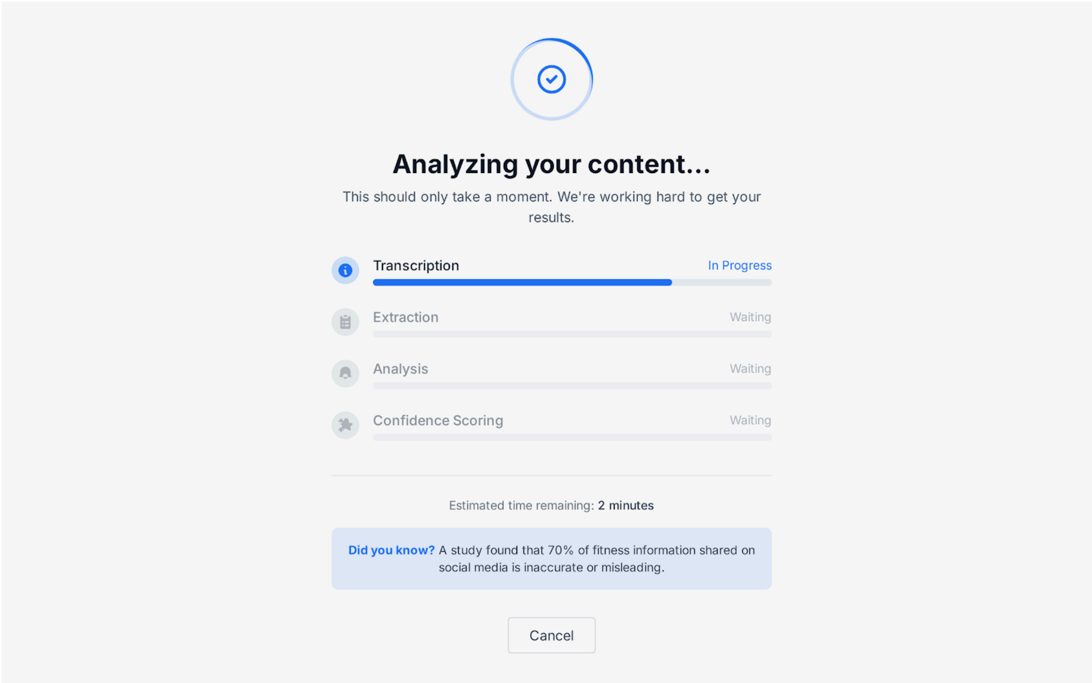
Analyzing your content...
This should only take a moment. We're working hard to get your results.
Transcription
In Progress
Extraction
Waiting
Analysis
Waiting
Confidence Scoring
Waiting
Estimated time remaining: 2 minutes
Did you know? A study found that 70% of fitness information shared on social media is inaccurate or misleading.
Cancel

Figure 8.

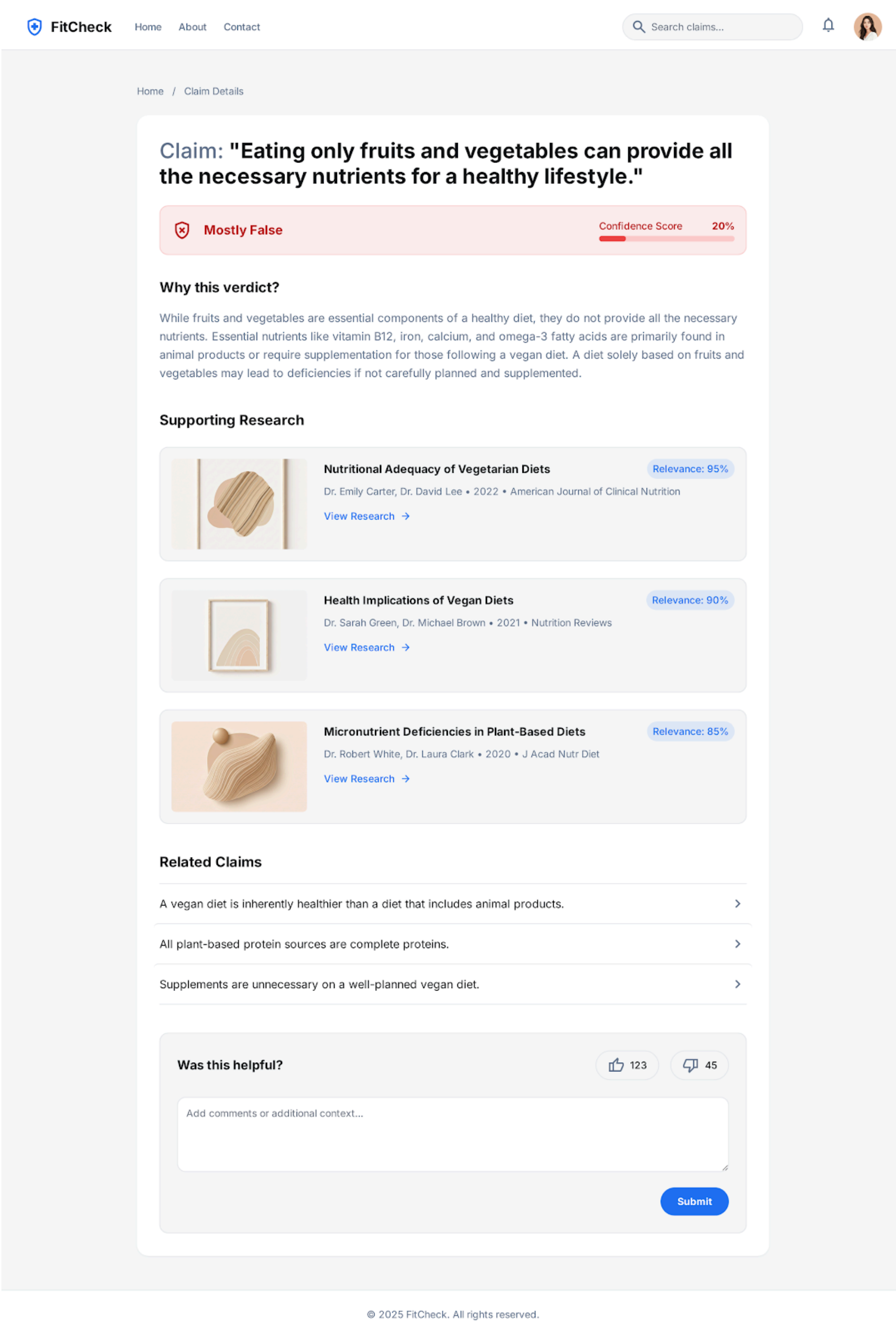
FitCheck
Home
About
Contact
Search claims...
Home / Claim Details
Claim: "Eating only fruits and vegetables can provide all the necessary nutrients for a healthy lifestyle."
Mostly False
Confidence Score
20%
Why this verdict?
While fruits and vegetables are essential components of a healthy diet, they do not provide all the necessary nutrients. Essential nutrients like vitamin B12, iron, calcium, and omega-3 fatty acids are primarily found in animal products or require supplementation for those following a vegan diet. A diet solely based on fruits and vegetables may lead to deficiencies if not carefully planned and supplemented.
Supporting Research
Nutritional Adequacy of Vegetarian Diets
Relevance: 95%
Dr. Emily Carter, Dr. David Lee • 2022 • American Journal of Clinical Nutrition
View Research
Health Implications of Vegan Diets
Relevance: 90%
Dr. Sarah Green, Dr. Michael Brown • 2021 • Nutrition Reviews
View Research
Micronutrient Deficiencies in Plant-Based Diets
Relevance: 85%
Dr. Robert White, Dr. Laura Clark • 2020 • J Acad Nutr Diet
View Research
Related Claims
A vegan diet is inherently healthier than a diet that includes animal products.
All plant-based protein sources are complete proteins.
Supplements are unnecessary on a well-planned vegan diet.
Was this helpful?
123
45
Add comments or additional context...
Submit
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Figure 9.

3.3.5 Applications

3.3.5.1 Use Case Diagram

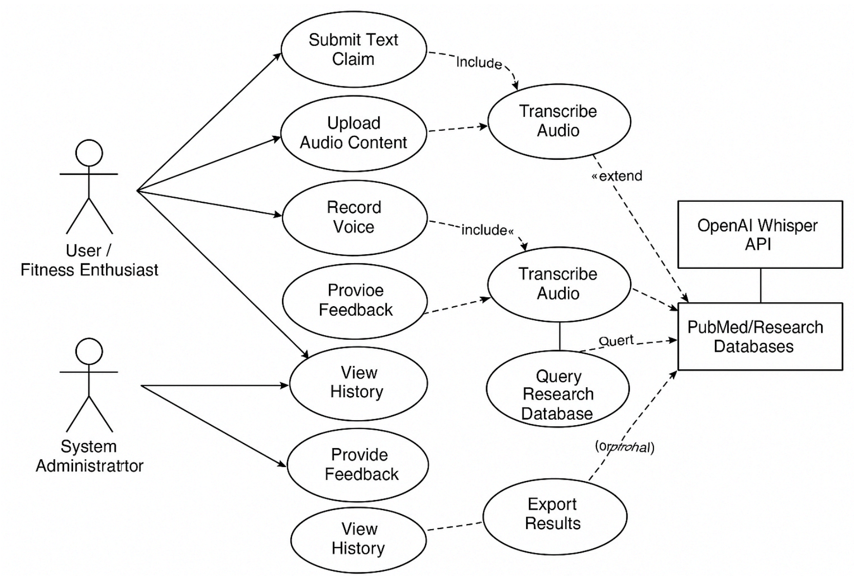

Figure 10.

3.3.5.2 User Flow Diagram

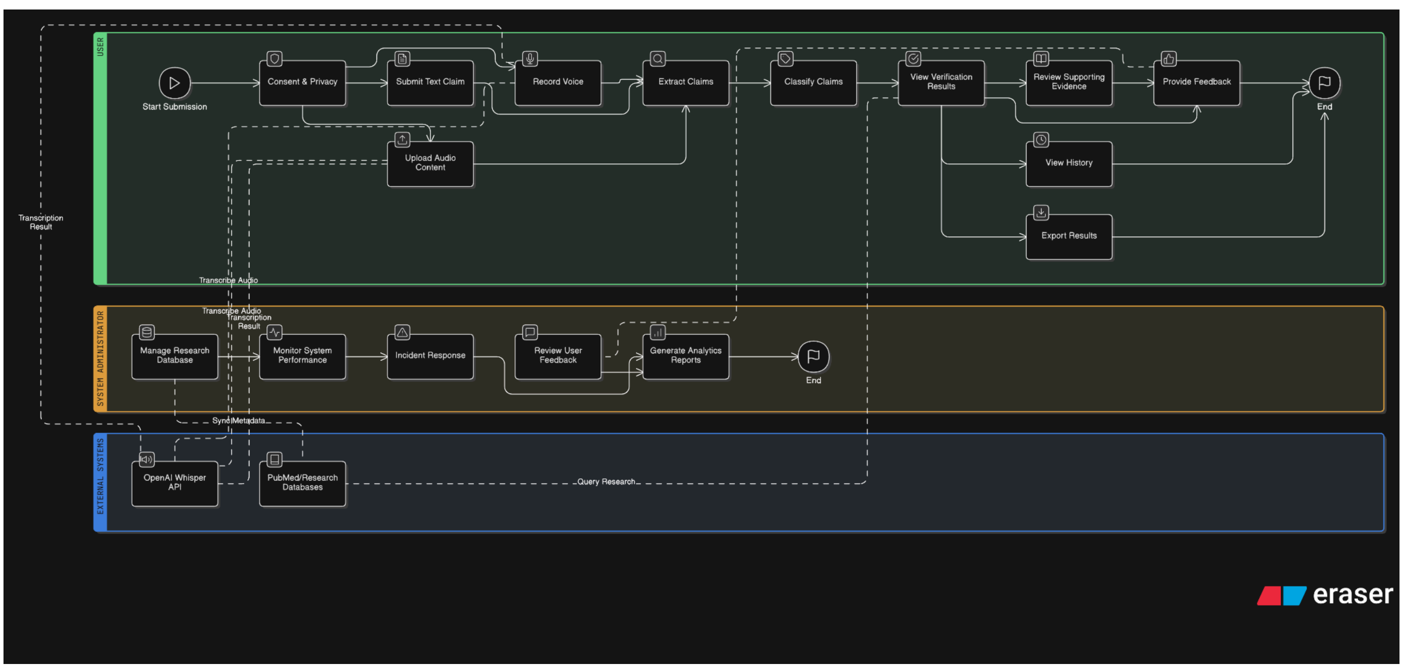

Figure 11.

3.3.6 Technology

This section defines the complete technology stack with justifications for each choice.

Development Technologies

Component Category	Technology/Service	Justification/Purpose
Presentation Layer (Frontend)	**React.js 18+**	Component-based architecture for reusable UI elements; efficient rendering for dynamic content.
Frontend State Management	**Redux Toolkit or Context API**	Centralized state management for user sessions, history, and verification results; aids debugging.
Backend Framework	**Node.js with Express.js 4.18+**	Non-blocking I/O for handling concurrent API requests; JavaScript full-stack consistency.
API Architecture	**RESTful API with JSON**	Industry standard; stateless architecture supports horizontal scaling and language-agnostic data exchange.
Natural Language Processing (NLP)	**Natural (Node.js library) & Compromise.js**	Provides tokenization, stemming, classification, and entity extraction; pure JavaScript implementation eliminates Python dependencies.
Machine Learning (Inference)	**TensorFlow.js or Brain.js**	Allows running ML models in Node.js; integrates seamlessly with the Node.js ecosystem.
Language Models	**Hugging Face Inference API or ONNX Runtime**	Access to transformer models (like BERT) for semantic similarity without Python dependencies.
Primary Database	**PostgreSQL 15+**	ACID compliance for data integrity; **Full-text search** capabilities essential for querying the research paper data.
Caching Layer	**Redis 7+**	In-memory store for caching frequent verification results; reduces database load and supports session management.

Database ORM/Query	**Prisma or Sequelize**	Simplifies database operations (e.g., migrations, seeding) and offers type-safe access.
External API (Audio)	**OpenAI Whisper API**	State-of-the-art accuracy for **speech-to-text** transcription; robust handling of various accents and audio quality.
External API (Research)	**PubMed/NCBI API** (Optional)	Free access to biomedical literature metadata (DOI, citation counts, abstract text) to enrich the curated database.
Cloud Hosting	**AWS or Google Cloud Platform (GCP)**	Provides compute instances, container orchestration (Docker), and load balancing for scalability.

Table 2.

3.3.7 Deployment

Infrastructure Technologies

Cloud Hosting Platform: The system will be deployed on Amazon Web Services (AWS) or Google Cloud Platform (GCP), selected based on cost-effectiveness during the development phase. Key infrastructure services include:

- Compute instances (AWS EC2 or GCP Compute Engine) with auto-scaling capabilities
- Container orchestration using Docker for consistent deployment environments
- Load balancing services for distributing traffic across multiple application instances
- Content Delivery Network (CDN) integration for static asset delivery

Development Environment: Local development will utilize:

- Operating System: Ubuntu 22.04 LTS or macOS for development machines
- Container Runtime: Docker Desktop for containerized development
- Version Control: Git with GitHub for repository hosting and collaboration
- IDE: Visual Studio Code with Python and JavaScript extensions

3.4 Resources

3.5 Project Schedule and Gantt Charts

Task No.	Task Name	Planned Hours	Planned Start Date	Planned End Date	Deliverable*
1	Research & Data Curation (Phase 1)	30	12/01/2025	12/10/2025	Research notes on 8-10 papers & abstracts
2	Research & Data Curation (Phase 2)	30	12/11/2025	12/20/2025	500+ research papers collected and processed
3	Setup & Schema (Database & Backend)	30	12/21/2025	12/31/2025	Postgres database implemented with sample data & Flask API basic endpoints
4	NLP Core Development	30	01/01/2026	01/10/2026	Text preprocessing and Entity Recognition modules tested
5	Claim Extraction & Whisper API Integration	30	01/11/2026	01/20/2026	Claim extraction pipeline and audio transcription functionality
6	ML Training Data Collection & Labeling	35	01/21/2026	01/31/2026	300+ labeled claims for model training
7	Train, Validate, & Tune ML Model	40	02/01/2026	02/10/2026	Validation report showing 75%+ accuracy
8	Implement Verification Engine (Scoring & Queries)	35	02/11/2026	02/20/2026	Complete verification system with confidence scoring and relevance ranking

9	Frontend Development (Setup & Input Pages)	30	02/21/2026	03/02/2026	React scaffold, routing, and home/input screens
10	Frontend (Results Display & User Features) & Integration	35	03/03/2026	03/10/2026	Results dashboard and user authentication/history features connected to API
11	User Testing & Bug Fixes	35	03/11/2026	03/14/2026	Testing report with user feedback and bug fix log
12	Final Deployment & Documentation	30	03/15/2026	03/18/2026	Live application URL and complete final project report (Chapters 1-6

Table 3.

3.6 Project Gantt chart

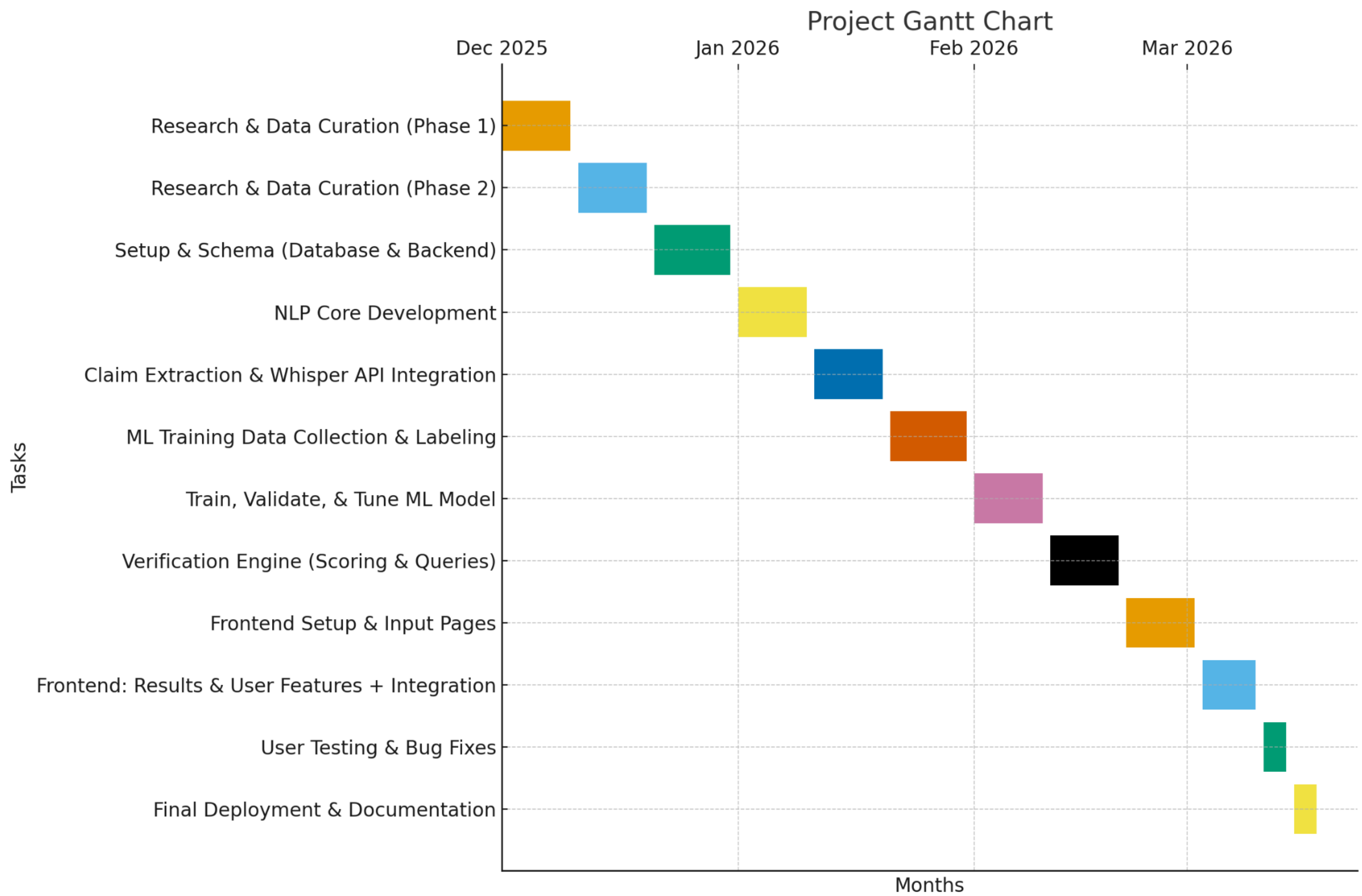
Project Gantt Chart
Dec 2025
Jan 2026
Feb 2026
Mar 2026
Research & Data Curation (Phase 1)
Research & Data Curation (Phase 2)
Setup & Schema (Database & Backend)
NLP Core Development
Claim Extraction & Whisper API Integration
ML Training Data Collection & Labeling
Train, Validate, & Tune ML Model
Verification Engine (Scoring & Queries)
Frontend Setup & Input Pages
Frontend: Results & User Features + Integration
User Testing & Bug Fixes
Final Deployment & Documentation
Tasks
Months

Figure 12.

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